

ELECTRIC TWO-WHEELER MARKET ANALYSIS

A Power BI Dashboard on India's EV Two-Wheeler Market

Capstone Project Report | Tool: Microsoft Power BI

15 EV Models	4 Brands	2M Monthly Sales	122.42 km Avg Real Range	4.24 / 5 Avg Rating
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1. Introduction

Electric two-wheelers are becoming a popular choice in India as more people look for cheaper and cleaner ways to travel. Brands like Ola, TVS, Bajaj, and Ather now sell many different electric scooter models, each with its own range, speed, price, and rating. With so many options, it becomes hard for buyers and companies to compare them all at once.

For this capstone project, I built a Power BI dashboard using electric two-wheeler data. It covers 15 models from 4 brands, with total monthly sales of about 2 million units, an average real-world range of 122.42 km, and an average customer rating of 4.24 out of 5. The goal was to turn this raw data into simple charts that show which brands and models actually perform best.

2. Challenge the Company Was Facing

EV companies and dealers collect a lot of data on sales, range, speed, and ratings, but without a clear report, they face several problems:

- They did not know which states were adopting EVs the fastest, making expansion planning difficult.
- It was hard to compare brands fairly on range, charging speed, and price.
- They did not know which price segment (budget, mid-range, premium) sells the most.
- There was no clear view of which models actually give the best value for money.
- Range, speed, rating, and sales data were spread across many separate sheets.

3. Data Preprocessing Challenges I Faced

Before building the dashboard, I had to clean the raw data. This step took real effort because the data had several issues:

- **Duplicate rows:** Some EV models appeared more than once in the data. I had to find and remove these duplicates so sales and range numbers were not counted twice.
- **Missing values:** A few models were missing their rating or top speed values. I checked each case carefully instead of deleting rows, so I would not lose useful data.
- **Inconsistent naming:** Model and state names were written in different ways (e.g. "iQube ST" vs "iqube-st", "UP" vs "Uttar Pradesh"). I had to standardise these names so they would group and map correctly.

- **Wrong data types:** Columns like battery capacity, range, and price were sometimes stored as text. I had to convert them to numbers before doing any calculations.
- **Outliers:** A few models had real-world range values that looked unrealistic compared to their battery capacity. I checked and corrected these so they would not distort the charts.
- **Formatting sales numbers:** Monthly sales figures were written in different formats (some with commas, some in short form like "0.6M"). I converted all of them into one consistent number format.

Cleaning the data first was essential. Without this step, the map, rankings, and averages on the dashboard would have been misleading.

4. The Approach I Decided to Take

Once the data was clean, I planned the dashboard around real questions a buyer or company would ask:

- Studied all columns: brand, model, state, battery capacity, range, top speed, price segment, rating, and monthly sales.
- Grouped and standardised the data so brand and state comparisons were fair.
- Designed each chart to answer one clear question, like "Which brand gives the best value for money?"
- Used a map to show which states are adopting EVs the fastest.
- Kept the design simple with KPI cards, bar charts, line charts, and donut charts.

5. Highlights of the Important Steps

- **Overview page:** Added KPI cards for total models, brands, monthly sales, average range, and average rating, plus a map showing which states are adopting EVs the fastest. Maharashtra and Karnataka lead by a clear margin.
- **Range vs battery capacity:** Built a chart showing that real-world range increases steadily as battery capacity goes up, from about 75 km at 2 kWh to over 180 km at 4.5 kWh.
- **Charging and range by brand:** Compared brands on fastest charging and longest range. TVS came out on top, ahead of Ather, Ola, and Bajaj.
- **Sales and ratings by brand:** Found that Ola leads in monthly sales, while TVS actually has the highest customer ratings, followed closely by Ather, Ola, and Bajaj.
- **Value for money:** Scored each brand on range, rating, and price together. Ola came out as the best value for money, followed by TVS, Bajaj, and Ather.
- **Price segment analysis:** Compared sales across budget, mid-range, and premium segments. Mid-range vehicles sell far more than premium or budget models.
- **Top speed and range by model:** Ranked individual models by top speed and real-world range. The S1 Pro is the fastest model, while the iQube ST has the longest real-world range.
- **Market share:** Found that Ola holds more than half the market (53.84%), followed by TVS, Bajaj, and Ather.

6. How This Will Help the Client Once Deployed

- **Smarter expansion:** The company can focus on states like Maharashtra and Karnataka first, where EV adoption is already growing fastest.
- **Better product focus:** Since mid-range models sell the most, the company can prioritise building and marketing more mid-range options.
- **Stronger competition on quality:** Knowing that TVS leads in ratings and charging speed, other brands can improve these areas to compete more effectively.
- **Clearer marketing:** Models with the highest range or top speed, like the iQube ST and S1 Pro, can be highlighted directly in ads and campaigns.
- **Faster decisions:** Managers can check the live dashboard anytime instead of waiting for a report to compare brands and models.

In short, this dashboard turns raw EV data into a simple tool that helps the company plan expansion, guide product focus, and market smarter.

7. Conclusion

This dashboard gives a clear picture of India's electric two-wheeler market, and a few patterns stand out. Ola is the clear market leader, holding more than half the market share and also scoring highest on value for money, which explains why it sells the most units. However, Ola does not win on every measure: TVS actually leads in customer ratings and offers the fastest charging with the longest range, showing that Ola's market lead comes more from price and value than from having the single best product.

Maharashtra and Karnataka are adopting EVs the fastest, well ahead of other states, which points to strong demand in these urban and semi-urban markets. Mid-range vehicles clearly dominate sales over premium and budget options, showing that most buyers want a balance of price and performance rather than the cheapest or most expensive choice. Finally, performance leadership is split across models rather than one single brand: the S1 Pro leads on top speed, while the iQube ST leads on real-world range, and real-world range itself rises steadily with battery capacity, confirming that battery size remains the strongest driver of how far these vehicles can travel.